

SIM WENG JIN

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PROFILE

AI Engineer with experience developing AI agent chatbot systems for SaaS-based customer support and business operations. Currently working on a live chat platform where AI agents answer customer enquiries, call tools, trigger automation workflows, integrate with backend systems, and hand off conversations to human agents.

PROFESSIONAL EXPERIENCE

Des Digital Marketing Sdn. Bhd

Apr 2025 – Present

AI Engineer

- Develop and maintain an AI agent chatbot system for a SaaS-based live chat and customer support platform, enabling AI-assisted customer enquiries, tool calling, automation workflows, and human agent handoff.
- Integrated AI chatbot capabilities with backend systems to support business-specific responses, customer support workflows, and external system connectivity.
- Built and maintained live chat features including customer conversations, operator support flow, realtime message handling, and AI-to-human escalation.
- Designed AI agent workflows that allow the chatbot to perform actions, trigger automated processes, and support customer service operations beyond basic question answering.

Infineon Technologies (M) Sdn. Bhd

Oct 2024 – Mar 2025

AI Intern

- Developed a federated learning system using NVFlare to support collaborative model training across decentralized devices.
- Converted centralized machine learning workflows into a federated learning setup for distributed training experimentation.
- Configured industrial cameras and Jetson AGX Orin edge devices for local image collection, processing, and testing.
- Developed and optimized LLM prompts for Failure Modes and Effects Analysis (FMEA) tasks.
- Performed image annotation and supported troubleshooting for computer vision and edge AI projects.

EDUCATION

BACHELOR OF COMPUTER SCIENCE

Oct 2021 – Mar 2025

UNIVERSITI TEKNIKAL MALAYSIA MELAKA (UTeM)

- Major in Artificial Intelligence.
- CGPA: 3.71
- Thesis on "Interactive Feedback System: Implementing Gesture Recognition for Enhanced User Engagement"

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL, Dart

AI / LLM / Agent Development: LangChain, LangGraph, LlamaIndex, Prompt Engineering, Tool Calling, AI Agent Workflow, RAG

Backend Development: FastAPI, Django, REST API, WebSocket, SQLAlchemy, Pydantic, Celery, Redis

Frontend Development: React, Vite, HTML, CSS

DevOps: Docker, Docker Compose, Nginx, PgBouncer, Vercel, Prometheus, Grafana

Operating Systems: macOS, Windows, Linux Ubuntu

SPECIALIZATION

AI Agent Development, Tool Calling, Workflow Automation, LLM Applications, Retrieval-Augmented Generation (RAG), SaaS-based Customer Support Automation, Federated Learning, Computer Vision

LANGUAGES

Mandarin (Native), English, Malay

PROJECTS

Group Chat AI Agent

In Development

- Developing an AI assistant for group conversations that responds when mentioned, supports task-based actions, and is designed for future integration with multiple messaging platforms.

Ransomware Detection System | Project Helper

Mar 2026 – Apr 2026

- Developed a machine learning-based detection system to classify suspicious file or system behaviour patterns for cybersecurity experimentation.

Expense Tracker App with OCR-based Receipt Scanning**Jan 2026 – Apr 2026**

- Developed a personal finance tracking app that uses OCR to scan receipts, extract transaction details, categorize expenses, and reduce manual data entry.

Large Language Model FMEA (RAG)**Oct 2024 – Mar 2025**

- Developed a RAG-based chatbot to support Failure Modes and Effects Analysis (FMEA) by retrieving relevant knowledge and generating context-aware responses.

Federated Learning with NVFlare Framework**Oct 2024 – Mar 2025**

- Built a federated learning workflow using NVFlare to support decentralized model training across edge devices and local data sources.

Interactive Feedback System: Implementing Gesture Recognition for Enhanced User Engagement**Mar 2024 – Aug 2024**

- Developed a computer vision system that allows users to answer survey questions using hand gestures, with realtime recognition and survey result storage.